Trends in
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Feature Review

Is the impact of spontaneous movements on early visual cortex species specific?

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Recent studies in non-human primates do not find pronounced signals related to the animal's own body movements in the responses of neurons in the visual cortex. This is notable because such pronounced signals have been widely observed in the visual cortex of mice. Here, we discuss factors that may contribute to the differences observed between species, such as state, slow neural drift, eccentricity, and changes in retinal input. The interpretation of movement-related signals in the visual cortex also exemplifies the challenge of identifying the sources of correlated variables. Dissecting these sources is central for understanding the functional roles of movement-related signals. We suggest a functional classification of the possible sources, aimed at facilitating cross-species comparative approaches to studying the neural mechanisms of vision during natural behavior.

Highlights

Studies in non-human primates observed only small modulations by spontaneous body movements in early visual cortex, contrasting with the observations of substantial movement-related effects in mice.

Disentangling the sources of the movement-related neural activity in the visual cortex is challenging because many physiological and behavioral factors co-vary.

Categorizing these sources, ranging from more sensory to more motor or action-related factors, may help resolve apparent differences in findings across species.

While some discrepancies may reflect analysis or experimental factors, common principles in how movements modulate visual processing across species may emerge, especially when considering processing of the central versus peripheral visual field.

Movement-related neural activity in visual cortex differs between mice and monkeys

How the brain processes visual information during active behavior and gives rise to visual perception is a longstanding question in neuroscience [1–3]. In 2010, it was reported that neural activity in the mouse primary visual cortex (V1) increased strongly when a mouse was running or walking compared with when it was sitting still [4] (see also [5–9]). More recently, a series of studies [10–12] showed that, even beyond the effects of running, spontaneous body movements, mainly of the face, were also associated with substantial modulation of neural activity in the visual cortex. Intriguingly, some of these findings persisted when the mice were in the dark [10], suggesting that they reflected non-retinal factors.

These findings in mice posed a challenge to the textbook view of how the visual cortex works. While early work in cats and macaques observed some response modulation with wakefulness [13] or eye movements [14], visual responses in V1 are thought to mainly depend on the visual stimulus in the receptive field and its surround. Had previous work in mammals other than mice (mostly cats and primates) missed such substantial response modulations because body movements were not monitored or do modulations by spontaneous body movements reflect species differences between the visual cortex in mice and primates? Recent findings in primates provide data that may help resolve this puzzle (Figure 1).

One study examined the effect of locomotion in marmosets under conditions carefully matched to those of experiments performed in mice [15]. Another study explored the effect of spontaneous body movements other than locomotion in macaques [16]. Both studies sought to match the data analysis methods to those in mice. In the marmosets, running was associated with a small decrease in gain, compared with the substantial gain increase found in mice (Figure 1A,B). In a recent follow-up preprint [17], a significant increase in gain was reported for more eccentric receptive fields in marmoset V1. Although still smaller than in mice, this gain was in the same

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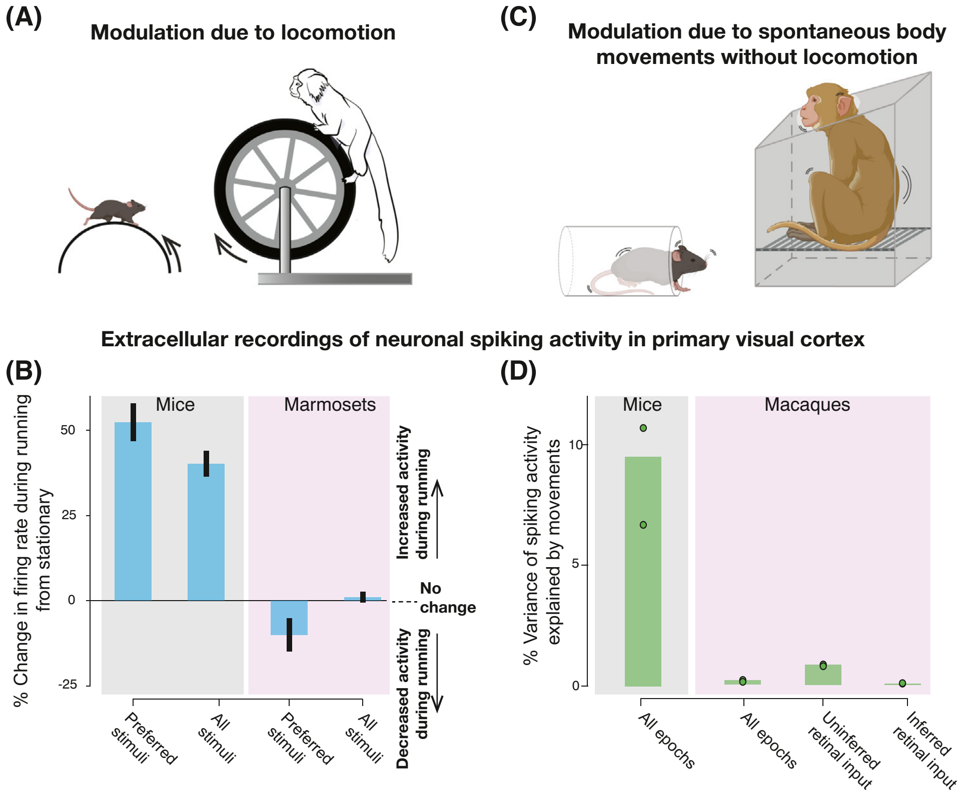

Figure 1. The effects of movement in early visual cortex differ between mice and non-human primates. (A,B) The impact of locomotion in mice and marmosets [15]. (A) Illustration of the experimental setup. The study used an openly available data set of extracellular recordings from head-fixed locomoting mice, and adapted a similar experimental setup to record in head-fixed locomoting marmosets. (B) Locomotion increased the gain in neuronal responses to preferred visual stimuli by over 50% in mice, but resulted in a much smaller, and suppressive, effect in marmoset foveal primary visual cortex (V1). Bars indicate mean across all recorded units across animals; vertical lines represent 95% confidence intervals. (C,D) The impact of spontaneous body and facial movements in non-locomoting animals for mice [11] and macaques [16]. (C) Illustration of the experimental setups. Both studies used head-fixed animals performing visual tasks while monitoring spontaneous body movements using videography. (D) The amount of variance in spiking activity that was explained by spontaneous movements was about an order of magnitude higher in mice than in primates. In the primate, data were analyzed separately when the input to the receptive field (RF) of individual units could be inferred (i.e., when the RF was within the display screen) versus not, showing that the modulation could be accounted for almost completely by changes in the inputs to the RF during movements rather than by movements themselves. Bars indicate mean across animals; data points indicate mean across units for each animal. Adapted from Figure 3 in [15] (B), and Figure 8E in [11] and Figures S5A and 2f in [16] (D). Figure partially created using BioRender (biorender.com).

direction as that observed in mice. As discussed in later sections, differences between central and peripheral vision in the context of movement-related signals require further investigation. In the macaques (up to 14 degrees eccentricity; Figure 1C,D), the modest amount of variance explained by spontaneous body movements disappeared once movements that changed the retinal input were accounted for (Box 1). Thus, both studies [16,17] found that the pronounced modulatory effects associated with movements observed earlier in mice were substantially reduced or nearly absent in non-human primates(Figure 1). These findings mirrored the overall small and mixed effects of body movements on visual processing in humans [18–20].

To address these discrepancies between studies, we first propose five categories of effects that may all manifest as movement-related modulations of neural activity in the visual cortex. This

Box 1. Experimental control versus measuring and accounting for covariates

Studies in neuroscience typically aim to control for most variables to study one variable of interest. However, 'control for' can either mean direct 'experimental control' of variables, or allowing them to vary, measuring them and then 'accounting for' them in subsequent analysis. These represent two distinct approaches, and the use of the terms 'control for' has evolved over time, particularly with the move toward more unconstrained and naturalistic behavior, as well as advances in analysis methods.

Experimental control in the context of visual neuroscience generally implies restricting movements, such as head and gaze fixation, and presenting defined stimuli on a monitor. This direct control makes it possible to readily dissociate potential confounds, such as changes in retinal input, and associate neural responses with the visual stimulus or other tasks variables. However, it can be challenging to achieve this level of experimental control, particularly in mice and in naturalistic behaviors that require direct interaction with the environment. Moreover, some degree of movement variability, such as microsaccades, is typically present even when subjects are trained to maintain visual fixation.

Thus, an alternative approach is achieved by measuring potential covariates, and accounting for them post hoc. A simple form of this involves restricting analyses to epochs when behavioral variables are matched, such as only analyzing stationary periods. A more general approach is current model-based methods that allow the impact of known covariates on visual processing to be quantified from single-trial and continuous data [11,16,23]. Importantly, relaxing behavioral control can allow for the discovery of novel effects, such as the initial findings on movement signals in the visual cortex of mice. However, for both approaches, the relevant variables need to be measured, but how do we know which variables are relevant? Indeed, one might not have considered locomotion as potentially relevant covariate until recent findings and, in some cases, eye movements are still not directly considered. Measuring as many covariates as possible and prioritizing the simplest possible explanation are good guiding principles (*c.f.* [138]). For example, in the context of understanding the role of visual cortex, given the powerful impact of retinal activity on V1, retinal input as a covariate should be prioritized. Additionally, a fruitful approach can be to infer latent variables directly from population recordings [139] that capture the statistical structure of the neural population activity, and seek corresponding behavioral covariates (e.g., [15,140]).

categorization seeks to clarify discussions around the effects observed across species and facilitate comparative approaches in visual neuroscience. Second, we discuss potential sources of the apparent species differences between mice and monkeys, focusing on movement-related signals in the early visual cortex, primarily V1.

Categorizing the potential sources of modulatory effects of spontaneous behavior in visual areas

Dissecting the sources of movement-related effects in visual areas exemplifies the general challenge of dissociating the underlying causes when the observable variables are correlated (Box 1). We suggest five categories (Figure 2), based on functional criteria, ranging from more sensory to more motor effects, which can result in movement-related activity in visual cortex.

Changes in feed-forward retinal input (also 'reafference' or 'visual feedback')

An obvious influence of movement on neural activity is that movement changes the visual input itself, by shifting the head and eyes relative to the environment. This effect of the actions of an animal on the sensory input was termed 'reafference', in contrast to changes in the environment itself ('exafference') or internal representations of the motor command ('corollary discharge/efference copy,' see 'Signals related to eye-movement and eye position'). Indeed, it is challenging to study movement without some amount of accompanying change in sensory input, and reafference may contribute to some findings to date.

Since changes in sensory input are an expected, or even trivial, consequence of movement, it is important to distinguish them when determining internally generated movement signals. One approach to achieve this is to keep the retinal input constant by eliminating periods of eye and head movement, either by fixing the gaze (e.g., 'control for eye movements'; Box 1) or analyzing only periods in which the eyes and head did not move. Another approach is to remove changes in visual input, either by darkness or a uniform gray screen. However, it is not trivial to remove all visible light, especially for nocturnal or dark-adapted animals. A more integrated approach is to

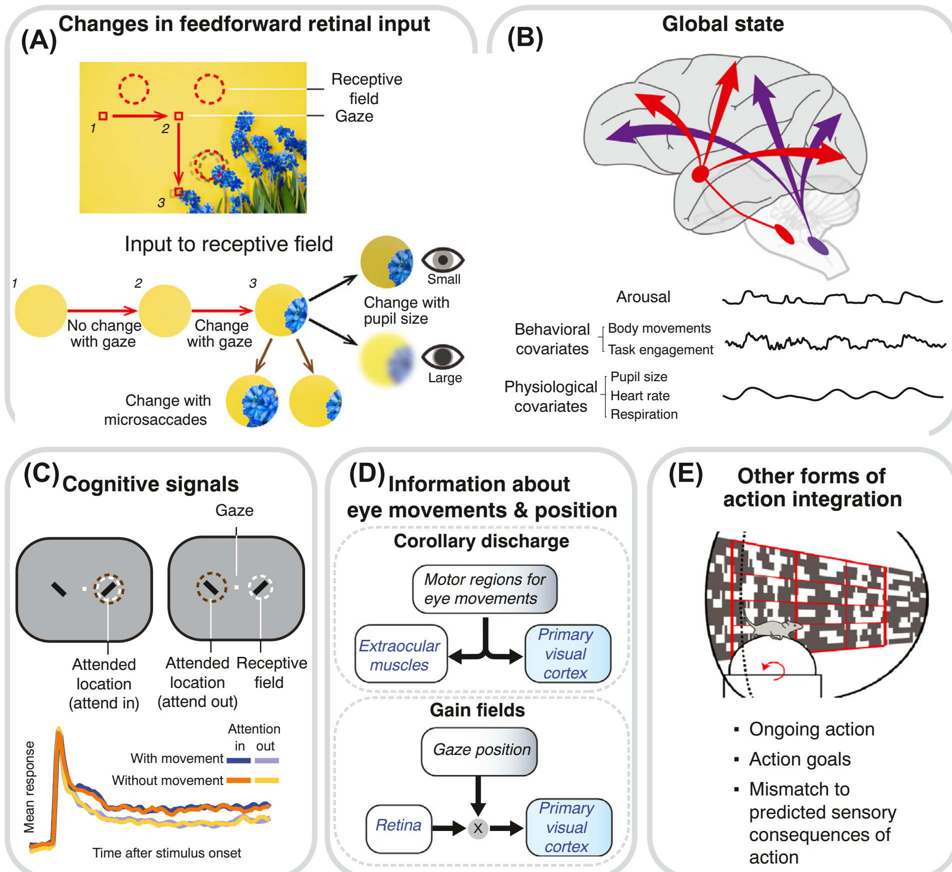

Figure 2. Potential sources of movement-related effects in early visual areas. (A) Changes in feedforward retinal input resulting from eye movements or pupil changes. If such changes in retinal input accompany body movements, they provide a possible explanation of movement-related modulation in early visual areas. (B) Changes in global brain state mediated by large-scale neuromodulatory projections, such as those of the cholinergic and noradrenergic systems. Fluctuations in global brain state can be coupled to behavioral variables (such as locomotion) or physiological variables (linked to arousal). (C) Changes in selective cognitive states, such as spatial attention, which may be correlated to behavioral output. While such effects may differ between studies, a recent study in macaques [16] found little support for this type of source. The neuronal responses for the same units are enhanced by spatial attention but not by the body movements of the animal. (D) Extra-retinal signals that provide information about movement and position of the eyes. These signals can help dissociate retinal effects due to movements of the eyes (A) from movements in the environment. (E) Other forms of action integration signals that convey specific information about self-movement can modulate visual processing based on current actions and their consequences/goals. Mean neuronal response timecourses adapted from Figure 3A in [16] (C).

determine the changes in sensory input and incorporate these into models, such as generalized linear models, of the neural responses by including both sensory and movement information as predictors (e.g., ‘account for eye movements’; Box 1 [21]). Although challenging, this is now possible in both head-fixed and freely gazing animals [22], and even in freely moving animals using head-mounted cameras [23,24].

While change in sensory input may be viewed as a nuisance variable when studying internally generated effects of movement and behavioral state, the impact of movement on the sensory input can be a meaningful component of visual processing during active vision. From the motor perspective, these changes provide sensory feedback on the outcome of a motor command.

From the visual perspective, movement through the 3D world shapes the visual input through effects such as optic flow [25], motion parallax [26], and dynamic occlusion [27], which can provide information about the structure of the visual scene. The structure of the neural activity may be informative in differentiating between theoretical accounts of the signal (e.g., [28]). Furthermore, eye movements format the visual input into a saccade-and-fixate cycle that shapes both the spatial and temporal statistics of the visual input [29]. Some impacts of movement on the retinal input are likely shared across species since they are fundamental aspects of analyzing the visual scene in natural contexts. Indeed, a recent study showed that, in both mouse and marmosets, saccadic gaze shifts induce a similar temporal sequence of visual processing in V1 [30].

Changes in general state such as arousal

Movement-related effects on neural activity in visual cortex can also result from changes in behavioral state associated with these movements, an effect well documented in mice. Recent excellent reviews presented general discussions of brain states [7,31,32], and we focus here primarily on movement-related changes in behavioral state.

Studies in mice showed that the movement-related changes in behavioral state manifest as increases in the gain of the neural responses in the visual cortex. The changes in behavioral state involve the basal forebrain cholinergic system [6,33,34], the pupil-linked cholinergic and noradrenergic arousal system [5,6,35], and thalamocortical interactions [36], and are mediated by specific local circuit elements [33]. Findings in mice also point to distinct modulatory influences by movements versus arousal-linked behavioral state. Behavioral state indexed by pupil diameter can modulate visual cortex in the absence of changes in observable behavior, such as whisking [36] and locomotion (e.g., reviewed in [7]). Modulations in arousal state accounted for seemingly multisensory effects in the visual cortex, such as responses to auditory stimuli [37] (but see also [38,39]).

Modulation by general arousal can have complex effects on visual responses. For example, if global state drives cholinergic or noradrenergic modulation that interacts with the local circuitry, it may change visual spatial integration [40]. In addition, acetylcholine can facilitate or suppress neuronal responses in V1 depending on its concentration (reviewed in [41]). Similar dose-dependent modulation linked to arousal [42] may account for the bandpass tuning to running speed observed in mouse V1 and dorsal lateral geniculate nucleus (dLGN) [5,9]. Moreover, global state changes can drive complex changes in tuning at the neural population level. For example, the gain modulation associated with locomotion shifts the neural population-level response to higher spatial frequencies, possibly supporting higher spatial acuity during locomotion [43]. However, this is not a result of changing the tuning of individual neurons, but of preferentially increasing the gain of neurons tuned to high spatial frequency [43]. Likewise, even a single gain signal can have different effects on distinct cell types, particularly as a result of inhibition and recurrent connectivity [44]. Thus, complex interactions with visual processing can result from unidimensional modulatory processes that are not directly 'multisensory' or 'sensorimotor'.

With respect to movement-related signals in the visual cortex, it is helpful to differentiate these effects from 'motor' signals reflecting the moving sensors (see 'Signals related to eye-movements and eye-position'), and mechanism of 'action integration' (see 'Signals reflecting action integration beyond movements of the sensors'). Modulation by movement-related behavioral state in mice has also been compared with the modulation by attention in primates (e.g., [45]). While there are parallels, it is useful to consider the interaction of these types of modulation with movement-related effects separately (see 'Signals relating to specific cognitive states such as selective attention'). The relevant difference for this purpose, particularly in primates, is that attention influences visual processing selectively depending on the task-relevant visual feature. Conversely, state changes, such as general arousal, act broadly [46,47]. Relatedly, both cholinergic modulation [48] and locus coeruleus-dependent noradrenergic modulation [49] are recruited in selective spatial attention in primates, in addition to their involvement in general, nonselective arousal.

Signals relating to specific cognitive states, such as selective attention

Modulation associated with cognitive signals, such as selective attention, decisions, or working memory, have been observed in the primate visual cortex, including modest effects in V1 [50–53], as well as in mouse V1 [54,55]. If idiosyncratic movements are systematically associated with these cognitive states, either source could masquerade as the other. Several reasons make it unlikely that cognitive effects can be fully explained by movement-related modulations. First, many cognitive effects are selective. For example, spatial attention is specific to neurons the receptive fields of which overlap with the attended spatial location. Feature-based attention or decision-related signals in feature discrimination tasks systematically depend on the feature selectivity of the neurons [51,56]. Thus, it would require nonuniform, feature-selective modulation by movements. Moreover, the modulation would have to systematically change with the attended spatial location, attended feature, or choice. Second, systematic differences in eye movements, such as fixational drift or microsaccades, associated with task demands, have often been accounted for in studies reporting modulation by cognitive factors (e.g., [57]). Finally, recent work [16] found little support for such association (see 'Do signals relating to specific cognitive states such as selective attention contribute to the differences in movement-related activity between mice and monkeys?').

Of course, the challenge of distinguishing between the sources of correlated cognitive signals in the visual cortex is not unique to movement-related effects. It has been discussed in other contexts, such as distinguishing the relative contributions of reward and attention [58] or sources of decision-related signals [59].

Signals related to eye movement and eye position

Movements that change the position of the eye have direct consequences on the visual input. Differentiating sensory input due to movements of the sensors from that of movements in the world is a critical function of nervous systems. In the visual cortex, three types of extra-retinal signal have been proposed to serve perceptual stability between eye movements and to transform the visual input from retinotopic coordinates to a spatial coordinate system on which the organism can act: (i) response modulation around saccadic eye movements; (ii) presaccadic remapping of the receptive fields; and (iii) gain modulation by gaze angle.

Corollary discharge is a copy of movement commands sent to sensory systems to disambiguate sensory input caused by self-generated movements [60]. These signals are used to prevent self-generated sensory input from triggering reflexes [61] or contaminating sensory input occurring in the environment [62]. During active sensing, animals use corollary discharge to create a sensory map and analyze objects in the environment [63,64]. In vision, corollary discharge has been implicated in driving response modulations around the time of saccades (saccadic modulation) and presaccadic remapping of receptive fields.

Saccadic modulation is typically a biphasic modulation of the spiking activity around the time of saccades, which has been observed in dLGN [65–70] and visual cortex, including V1, of both mice and macaques [14,21,65,66,69,71–79], even in the absence of changes in retinal input [21,66,67]. A recent study found that saccades suppressed macaque V1 responses to low spatial frequency stimuli, but enhanced responses to higher spatial frequencies, mirroring psychophysical effects in the same animals [78]. Inactivation studies in mice [66] and pigeons [65] suggest a motor origin of the signals (i.e., corollary discharge) driving saccadic modulation in

V1. Saccadic modulations may account for perceptual phenomena, such as saccadic suppression, and for perceptual continuity across saccades by modulating the sensitivity to the visual input around saccades [79].

Receptive field remapping, also thought to support perceptual continuity [79], refers to shifts of a visual receptive field immediately before a saccade to the corresponding space with respect to where the eyes will land after the saccade. In macaques, this was reported for extrastriate visual areas [80–85], but minimal for V1 [83]. Corollary discharge originating from the superior colliculus has been shown to be involved in receptive field remapping in the frontal eye fields in macaques [81]. In mice, instead of receptive field remapping, 'mismatch signals' have been described in visual cortex (see 'Signals reflecting action integration beyond movements of the sensors'), which may serve a similar computational role.

Modulation of visual responses by gaze angle was found in visual cortical areas, as early as V1, in mice, cats, and primates [23,86,87]. This modulation, often termed 'gain field', can be used to transform the retinotopic visual input to an egocentric spatial coordinate, (e.g., head- or body-centered reference frames) [88,89]. Thus, downstream areas could extract information about eye position without accessing the output of motor structures. Unlike saccadic modulations or receptive field remapping, evidence suggests that proprioceptive signals, rather than corollary discharge, are involved in creating gain fields [90].

Signals reflecting action integration beyond movements of the sensors

Information about self-movement may have a role in action integration, beyond accounting for movement of the sensors (eyes), as described in the previous sections. This has been demonstrated, for example, in mouse auditory cortex, where a specific inhibitory circuit based on input from motor cortex changes to cancel out the effect of self-generated sounds [91,92]. In the visual domain, studies in mice have shown that neurons in V1 encode mismatch when the visual feedback in closed-loop virtual reality is perturbed to no longer match the learned sensorimotor coupling [93–95]. The mismatch signals are spatially localized into mismatch receptive fields [94] and depend on inputs from anterior cingulate and motor cortices [94,95]. Furthermore, once the animal has learned the virtual environment, neurons represent expected visual features based on the current spatial location [96]. Together, these findings support a predictive coding model, wherein cortical neurons are continuously encoding the expected sensory consequences of current actions as well as deviations from the expectation [97]. Some components of the mismatch signals have been suggested to alternately represent feature encoding (optic flow contrast) combined with previously described locomotor gain effects [98,99], illustrating the difficulty in dissociating the various contributions.

What factors contribute to the observed differences in movement-related activity between monkeys and mice?

Cross-species comparisons of movement effects in visual cortex require consideration of the differences in their visual systems. We refer the reader to several reviews that address these comparisons [100–102], but begin this section with an overview of some critical considerations. We then discuss factors that may lead to differences in the modulation of visual responses by movement in mice and monkeys, using the categories discussed earlier. Finally, we address the importance of considering the peripheral versus central visual field.

Mice and monkeys share many aspects of the visual system, from the structure of the eye and central visual pathways to retinotopic organization and basic receptive field properties. However, notable specializations result in differences across the species. In primates, the primary projection of retinal ganglion cells (RGCs) is the dLGN of the thalamus, with <10% of RGCs projecting to superior colliculus and other targets [103]. Primates have color-encoding midget RGCs projecting exclusively to the dLGN, which have no direct counterpart in rodents. In rodents, only 25–50% of RGCs project to the dLGN, whereas most project to the superior colliculus (90%), with a significant number projecting to both [104]. Whereas the forward-facing eyes of the primate provide a mostly binocular view of the visual scene (about 120° horizontally), the lateral eyes of the mouse result in a reduced binocular zone (~40° horizontally). Global neuromodulatory systems, such as the cholinergic and adrenergic systems, project broadly to the visual cortex in both mice and primates, but differ at the level of the local circuit [105].

Primates have an ~100 times higher visual acuity compared with mice, and their vision is cone photoreceptor dominated, particularly in the fovea. The mouse retina is rod photoreceptor dominated, and lacks a clear fovea. However, it has been argued that mice have a specialized region covering ~30° of the visual field directly in front of the head, exhibiting increased density of specific types of RGC [106], increased cortical magnification, and slightly greater behavioral acuity [107]. This area has been termed the 'focea', and may be specialized for processing behaviorally relevant stimuli the animal is headed toward. Nonetheless, the decreased acuity and rod-dominated vision in mice suggest that their visual system functions similarly to that of the peripheral visual field in primates. This aligns with the putative preferential role of mouse vision in navigation and detecting salient stimuli over acuity-based tasks. Moreover, mice are primarily nocturnal, whereas primates are diurnal and, therefore, the two species are typically active at different ambient light levels. This results in overt specializations, such as the rod-dominated retina of the mouse, as well as circuit differences, such as dedicated circuitry that reduces locomotion at daytime light levels in mice [108]. This may also result in differences in the ethological consequences of coupling between movement and visual processing, which could confound direct comparisons.

As with other features of the visual system, eye movements in mice and monkeys have both shared and specialized characteristics. Primates make fast eye movements called saccades approximately three times each second on average, to center their high-acuity fovea on salient aspects of the scene. Mice make saccadic eye movements, but these are typically not independent of head movements [109,110]. Even in head-fixed mice, nearly all eye movements are coupled to attempted head movements [110]. This is consistent with the lack of a fovea, particularly since both the binocular zone and focea are larger than the range of eye movements, so that eye movements alone would not shift them onto a new region of the visual scene. Likewise, across the animal kingdom, there is an inverse relationship between the ratio of the mass of the head versus eye and the extent to which animals sample the scene by moving the head versus eyes [111]; in addition, active vision in mice is dominated by head movements more than in primates. Thus, it is possible that some mechanisms linked to eye movements in macaques are instead linked to head movements in mice. Indeed, both head orientation and eye position impact the gain of neuronal responses to visual stimuli in V1 of freely moving mice [23], similar to gain fields for eye position in macaques [87,112]. Furthermore, the tight coupling between head and eye movements in the mouse may imply that movement-related activity in head-fixed mice and that in head-fixed macaques, where saccadic eye movements can occur independently of head movement, are not directly equivalent. Head fixation disrupts the coupling of vestibular and corollary discharge signals with attempted head movements, which, given the forementioned considerations, could have a larger impact in mice. This concern is partly assuaged by studies with marmosets, which use rapid head movements to sample their visual scene more than macaques do [113,114], where modulation in the central visual field representation (~4°) due to locomotion is still an order of magnitude smaller than in mice [15].

Do changes in retinal input during movements contribute to the observed differences between mice and monkeys?

Body movements that are associated with movements of the retina, such as eye or head movements, will necessarily drive responses in the visual cortex, unless the retinal input is kept constant. Keeping the retinal input unchanged can be accomplished when the animal is in complete darkness (e.g., [10]), although removing all visible light is nontrivial, as discussed earlier. Moreover, changes in pupil size also change the retinal input, which can result in perceptual effects [115] and changes in feature selectivity in the mouse visual cortex [116]. Since movements are systematically linked with pupil size changes, studies have typically accounted for [8] or controlled (e.g., by pupil dilation [5]) for this pupil modulation. Disrupting the signal transmission from the retina to cortex pharmacologically has also been done to differentiate retinal and motor contributions of saccades [66]. Experiments examining the gain increase with locomotion in mice have addressed this by removing periods of eye movements, or comparing gain in the dLGN with that in the cortex, demonstrating that the overall gain increase with locomotion persists and is more pronounced in the cortex and, thus, not likely due to changes in retinal input [4,6,33].

In macaques, spontaneous facial movements appeared to modulate responses in the visual cortex, but these effects disappeared when the retinal input could be inferred and accounted for [16] (Figure 1). Relatedly, a recent study in freely moving mice demonstrated that apparent head-movement signals were dominated by the corresponding gaze-shifting eye movements and the resulting change in retinal input [30].

Since mice also make saccades, although less frequently when their head is fixed compared with primates (e.g., [117]), the presence of saccades may be a factor in some of the previous findings. The modulation by spontaneous body movements beyond locomotion in mice is largely explained by facial movements [10–12], which include the movements of the eyes. Work in mice reported that eye movements contributed to trial-to-trial neural variability, particularly in the visual cortex [12]. Therefore, it is possible that changes in retinal input partly contributed to the movement-related effects in mice, and that accounting for these would reduce the observed species difference.

Do differences in quantifying state modulations contribute to the differences in movement-related activity between mice and monkeys?

The locomotion-related modulations in mouse visual cortex predominantly increased the visually driven responses, with little change in spontaneous activity or stimulus selectivity, compatible with a gain change [4]. Subsequent work in mice identified more subtle effects beyond gain changes, such as changes to the spatial integration of visual stimuli [40]. Running-related effects in mice were identified even in the visual pathway upstream of V1, such as the dLGN [5] and retinal axons [118]. While effects of locomotion in mice are linked to changes in arousal, some findings supported distinct modulatory contributions by running and pupil-linked arousal (e.g., [8]).

Similarly, spontaneous body movements beyond locomotion can be correlated with behavioral state in mice. Unidimensional changes in behavioral state indexed by pupil diameter [10–12] also strongly correlate with modulations of neural activity in the visual cortex, even though multidimensional behavior quantified by videography still captured more neural variability.

The degree to which fluctuations in general arousal that may co-vary with spontaneous movements modulate activity in the macaque visual cortex is unclear. The study examining effects of spontaneous movements in macaque [16] sought to decouple effects of movements from effects of not only slowly changing variables, such as slow changes in behavioral state, but also slow nonstationary experimental factors, such as electrode drift (Box 2). Multivariate model-based

Box 2. Slow drift or nonstationarities complicate the measurement of correlation between behavior and neural activity

A concern in assessing whether behavioral measures and neural activity are correlated is the presence of slow changes in either signal (i.e., drifts). Regardless of whether these slow drifts are biological or reflect experimental instability, they pose major statistical challenges for the neuroscientist: because they were not randomized and each signal has substantial autocorrelation, it is easy to find 'nonsense correlations' between them [141].

To illustrate the challenge, it was shown using methodology typically used by the field that ~70% of mouse neurons display a false significant correlation with cryptocurrency, and Bonferroni correction only reduced the false positive rate to 35% [142]. Importantly, cross-validation within the session using withheld test sets did not protect against this false-positive rate. Of course, researchers in the field have long been rightly skeptical when findings rely on a wide distribution of correlation coefficients with zero mean, as was the case for the neural correlations with cryptocurrency. However, for widely used model-based analyses, such as generalized linear models, where beta coefficients can be either positive, implying positive correlations, or negative, implying negative correlations, this statistical challenge may be less obvious but is broadly equivalent. This suggests that any comparison between behavior and neural activity needs special attention to drift in firing rate, particularly when the behavior also shows slow fluctuations.

The potential for 'nonsense correlations', as discussed earlier, lurks in most large-scale neurophysiology data sets due to electrode drift. During a typical electrophysiology experiment, the electrode array will move independently of the neural tissue, causing instabilities in the recording. Although almost all spike sorting algorithms now implement some compensation for drift, this dramatically reduces spike-sorting quality even with ground truth knowledge of electrode movement [143].

Three proposals have been made for how to account for nonstationarity over long timescales. The first involves creating a proper 'null distribution', by pairing behavior and neural activity from separate sessions or by linearly shifting one of the signals [141]. Using this method reduces the false-positive rate from ~70% to 5% [142]. A second proposal is to explicitly include slow drift as a model parameter, and only interrogate significant correlations on smaller timescales [15,16,144]. A third proposal is to high-pass filter signals to remove low frequency components [10]. While none of these solutions fully resolves the issue, they represent emerging standard practices.

analysis allowed the authors to separate these factors. They implemented a set of covariates that accounted for slow fluctuations and spanned a few hundred seconds, thereby leaving spiking patterns at smaller timescales untouched. (Another approach is high-pass filtering the data, e.g., [9]; Box 2.) Spiking patterns at shorter timescales could be picked up by movement and task covariates. The analysis showed that, once retinal factors were accounted for (Figure 1D), movements did not explain activity in the recorded visual areas V1, V2, and V3/V3a [16]. Since slow fluctuations were captured as drift, it is unclear whether slow fluctuations in the state of the animal, such as general arousal, also modulated the neural activity. In future work, this may be addressed behaviorally, for example by manipulating task engagement. Of note, slow fluctuations in neural activity have been reported in area V4 of the macaque visual system [119].

Interestingly, analyses of recordings in marmoset V1 found little neuronal modulation with running movements, unlike in mice, although the analyses included slow fluctuations for both data sets [15] (Figure 1; note that slow drifts were accounted for in the additional population-based analysis in the study). This suggests that the discrepancy in the observed response modulation with movement between mice and monkeys is not explained by how the analyses treated slow drifts. At least in the settings of these experiments, body movements appeared to be more strongly associated with pronounced state changes in mice than in monkeys. Indeed, the coupling between behavioral states, such as locomotion, and general brain states, such as arousal, could vary greatly across species based on the behavioral repertoire of the species, making it challenging conceptually to define homologous states between species.

Do signals relating to specific cognitive states, such as selective attention, contribute to the differences in movement-related activity between mice and monkeys?

Recent work in macaques [16] directly compared modulation by spatial attention with modulation by spontaneous body movements (Figure 2C, bottom). The effect of movement was minimal and not correlated with the modulation by attention, suggesting that these processes were not coupled. Since modulation by movements in visual cortex is pronounced in mice, efforts to differentiate cognitive and movement sources of the modulations are particularly important [120–122] (Box 1). For example, choice encoding in the mouse visual cortex was no longer present once the visual stimulus and action were accounted for [122].

Do signals related to eye movement and eye position contribute to the differences in movement-related activity between mice and monkeys?

Saccadic modulations and gain fields have been observed in mouse V1 [23,66] and in primates [21,87]. If saccadic eye movements accompany body movements [123], saccadic modulations or visual response modulated by gaze angle could manifest as neuronal modulation by body movements. However, the neuronal modulation driven by the saccade itself appears small in the visual cortex, and instead reflects responses to the changed visual input due to the movements [30,78]. These modulations are tightly coupled to gaze shifts (i.e., a phasic response around each gaze shift), and differ from the temporal characteristics of the neuronal modulations with spontaneous body movements observed in mice.

Response modulations by gain fields also reflect interactions between gaze position and the retinal input. In natural environments, gaze shifts usually change the visual input. Thus, similar to saccadic modulations, the response of a neuron in visual cortex after gaze shifts will typically be a combination of the response to the new visual input (i.e., reafference or visual feedback) and its gain modulation due to gaze angle (Box 1). Unlike saccadic modulations, the temporal profiles of the modulations could resemble those by spontaneous body movements. Together, it appears unlikely that the overall small modulation of neural activity by eye movements or gaze position by themselves would explain the differences in modulations by body movements between mice and primates.

Do signals reflecting action integration beyond movements of the sensors contribute to the differences in movement-related activity between mice and monkeys?

Beyond binary locomotor state (running or not), many neurons in mouse V1 have been found to encode running speed, along with visual motion speed [9]. (Running speed encoding may partially reflect effects of arousal, see 'Changes in general state such as arousal'.) Effects of self-movement in virtual reality can manifest as spatial representations in V1, with neurons encoding the current location similar to place cells [120] and anticipated reward locations [121].

Additionally, neural signatures of a range of body movements have been observed in rodents. As discussed earlier, in head-fixed mice, task-related and spontaneous movements are correlated with neural activity [10,11]. In freely moving rats, neural activity across cortical areas reflects behavioral motifs, such as turning or rearing [124]. It appears reasonable that information about ongoing actions is broadcasted to relevant sensory areas, either to predict the impact of actions on the sensory input or to tailor sensory processing to intended actions and goals. Nonetheless, to our knowledge, a direct role of such mechanisms in visual processing has not yet been demonstrated. Furthermore, since many behavioral motifs can be related to eye movements, it remains to be determined how much of the neuronal modulation is due to change in retinal input.

These additional impacts of locomotion and body movement on neural activity have also not yet been observed in the visual cortex of primates to our knowledge. In some cases, such as the predictive coding in closed loop, the hypothesized mechanisms have not been tested, although visual response properties, such as surround suppression, have been explained within this framework [125].

Do movements primarily affect V1 neurons that represent the peripheral visual field?

Several of the previous studies in mice examining the modulatory role of movements (e.g., [4,10]) targeted the monocular zone of V1 with receptive fields at eccentricities of at least 20° [126]. By contrast, the findings in non-human primates were reported for receptive fields at smaller eccentricities. If modulation by movement differs for the central versus peripheral visual field representation, this discrepancy in eccentricity could contribute to the difference between the findings in mice and monkeys.

Indeed, in marmosets [15], a small suppression of the responses during running was found for neurons with parafoveal receptive fields (1–4° eccentricity). Conversely, for V1 units with more eccentric receptive fields (~20°), recorded in the calcarine sulcus, an enhancement of the responses by ~10% was observed during running, a relatively small effect compared with that observed in mice, but in the same direction. A recent preprint study in marmosets [17] reported that the modulatory effects increase even further, up to 30% for the far periphery. Nonetheless, in macaques [16], units recorded in the calcarine sulcus of V1 (eccentricities up to 14°) did not show more pronounced modulation by spontaneous movements compared with central units. However, it is possible that units representing the far peripheral visual field show stronger modulations, a question to address in future work (see Outstanding questions). Indeed, in macaques, the anatomical connections of feedback from downstream brain areas to the representation of central and peripheral V1 differ significantly [127]. For example, projections from auditory and multisensory areas preferentially target the peripheral compared with the central representation in V1 [127–129]. Such differential distribution of feedback projections could support a differential role of processing for the peripheral compared with the central visual field, with implications for spontaneous behavior and active vision. The representation of the far peripheral visual field in primates and most of the visual field in mice could primarily serve similar functions, relying on lower spatial acuity [100], such as supporting spatial navigation and orienting behavior, and, thus, share movement-related modulations.

As outlined in discussions earlier, analysis and experimental factors likely contributed to the pronounced differences in the modulation by movement observed between mice and monkeys. We expect that future analyses and experimental work will reveal the extent of the discrepancy between mice and monkeys when accounting for these factors (see Outstanding questions). Furthermore, the degree to which the modulation by movement depends on eccentricity could reflect a shared principle. Movements may mainly modulate low acuity visual processing, and, thus, the strength of neuronal modulation by movements may manifest specializations of how the different species use vision.

Concluding remarks

Movement-related neural activity in the visual cortex can result from many sources. These range from retinal effects that engage the relatively well-understood circuits from the retina to the visual cortex, modulations in general state, such as arousal, which are well characterized at the circuit level in mice but less so in monkeys, to modulations in cognitive state and mechanisms for action integration, which are complex and only partially understood.

When mice became more widely used for visual neuroscience, it was commonly emphasized that a key advantage of this animal model was the availability of genetic tools (e.g., [100]). However, using mice also encouraged the field to adopt more naturalistic behavioral paradigms and embrace the value (and challenges) of the variability of spontaneous behavior (Box 1). As these behavioral paradigms become more common in non-human primate research (*c.f.* [130]), it is beneficial to identify, classify, and scrutinize potential sources of species differences in visual

Outstanding questions

To what extent do changes in retinal input contribute to the sizable effects of spontaneous movements on visual responses in mice? Does the brain leverage the structure of the retinal input change resulting from the movements of the sensors (eye/head)?

To what extent is there a pure motor component of the movement modulatory signal in V1 that is not coupled to general state changes, such as arousal?

Are there fundamental differences in how movement and state changes (such as arousal) impact visual processing in mouse and primates, which may result from the distinct behavioral repertoire, ecological demands, and evolutionary history of the species? Are the states and movements being observed in current experiments truly comparable between mouse and monkey?

What are the movement-related signals in V1 used for? What is the impact of these signals in V1 on downstream visual processing and behavioral output?

To what extent do the signals reflect specific motor plans? Are there different timescales to these effects?

How do movement signals vary across the visual field, from central to peripheral vision? Is the primate central visual field the exception, in being specialized for high acuity image analysis, and free of movement modulation?

Is mouse V1 more comparable with higher visual areas or visuomotor areas in the macaque, and do movement signals have a more pronounced role in these higher areas in the primate?

neurophysiology. Different animal models have different advantages, and we believe that the field benefits from direct cross-species comparative research, including animal models other than the ones we focused on here, such as diurnal rodents, ferrets, tree-shrews, arthropods, and cephalopods (e.g., [30,131–137]). We aimed to provide a framework for such cross-species comparisons in the context of movement modulations. We presented a systematic delineation of the types of movement-related response in visual cortex that spans effects observed in rodents and primates, and outlined some of the remaining differences in how studies have been performed across these species (see Outstanding questions). We hope that continued investigation along these lines will help identify true differences, which may represent visual system specializations, as well as general principles that hold across species and reveal fundamental computations that are essential for active vision.

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Declaration of interests

The authors declare no competing interests.

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